

Loan Application Predictor

Team Grade A

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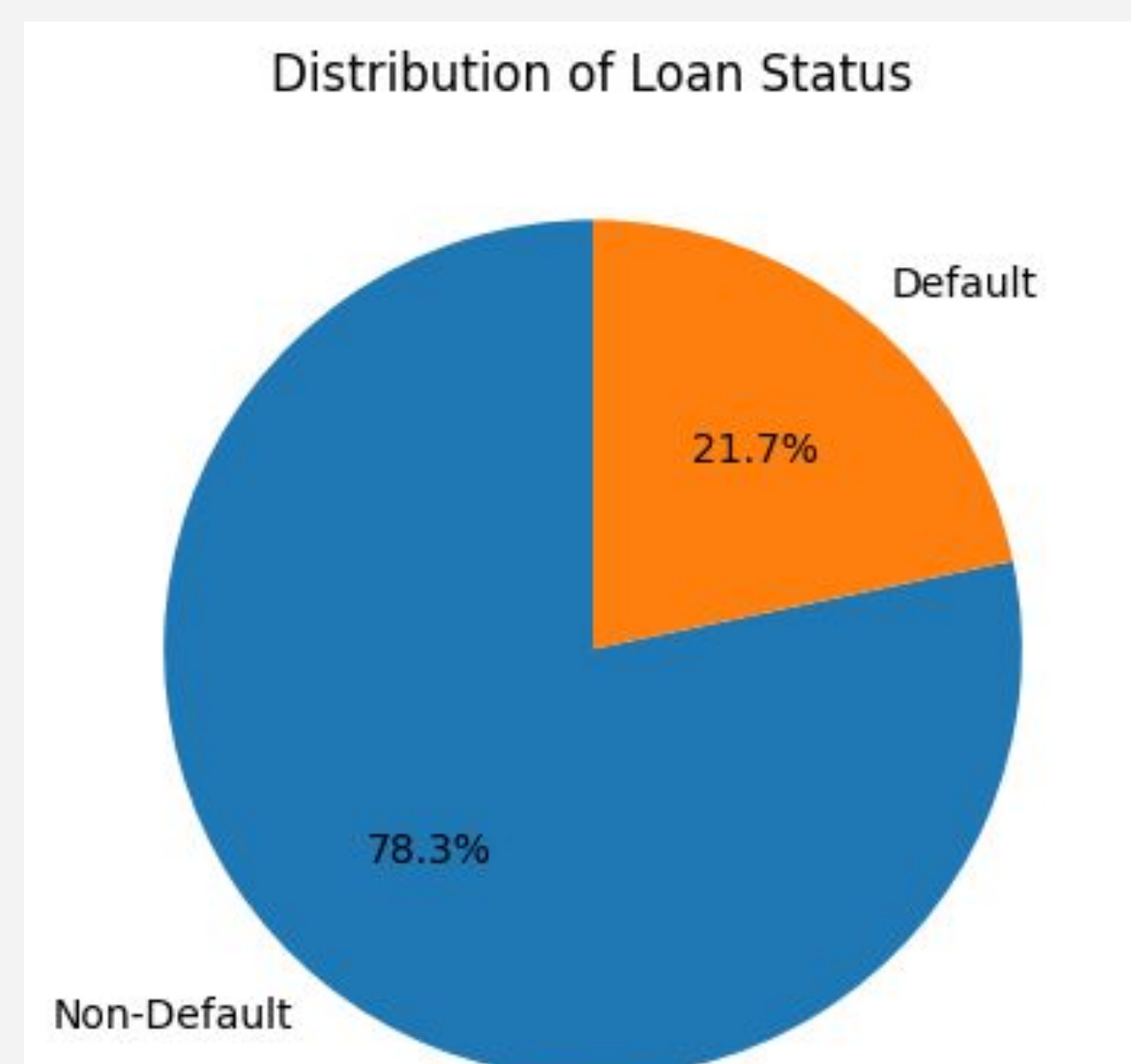
Data Science/AI Track

Using Logistic Regression and Random Forest Models we intend to help people understand their position when applying for a loan. This project is meant to take in user information and predict whether they would be accepted for a loan.

Background

Our data covers individuals between 18-84 and incomes \$4,000 and \$200,000 a year. Most borrowers either rent or have mortgages and these loans are used for a variety of reasons. In general, only around 21.7% default. Our dataset showed several variables that correlated with defaulting, these being the most significant:

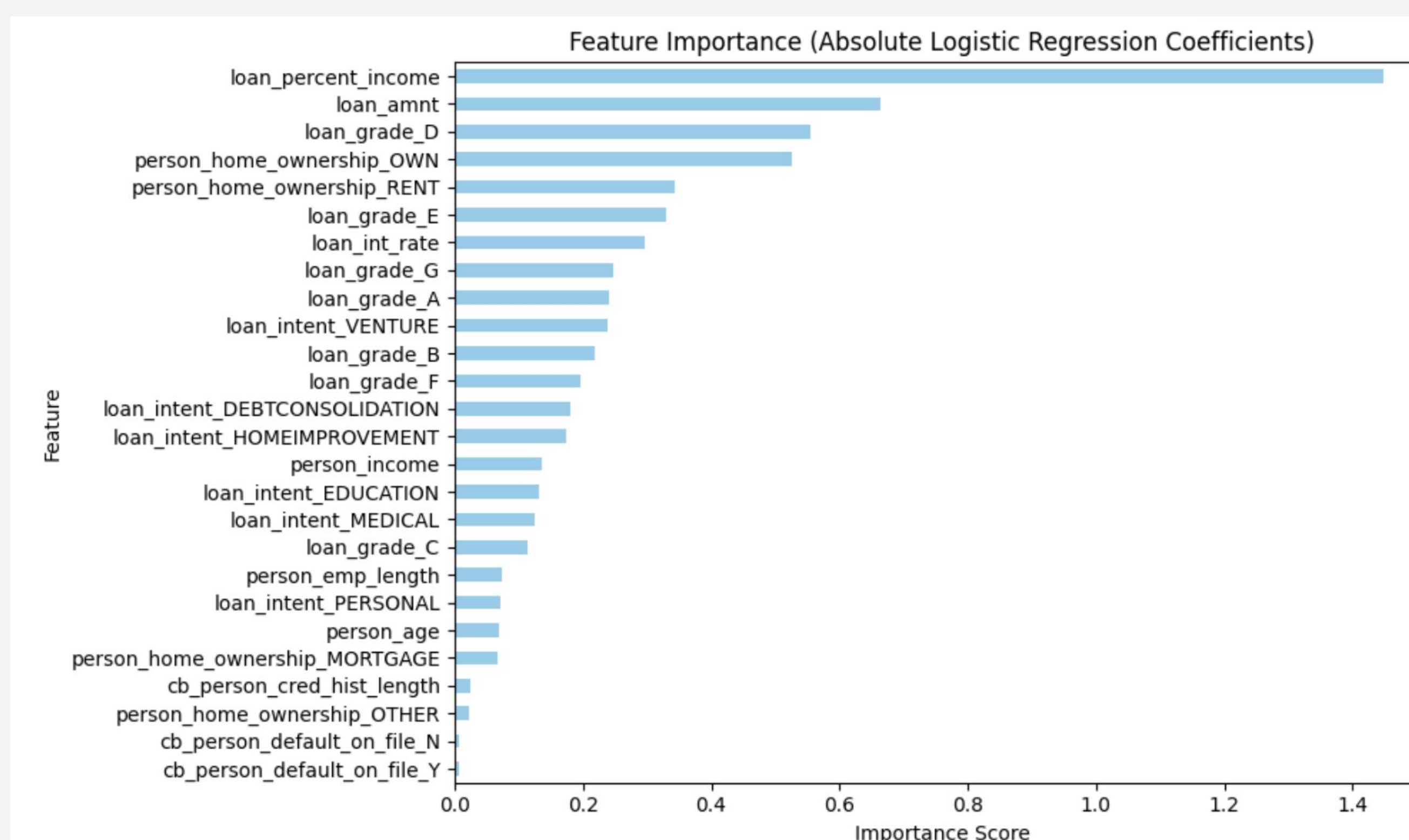
- High loan to income ratio
- High interest rates
- Poor Loan Grade
- Having a history of defaulting
- Not Owning a house or having a mortgage



Logistic Regression Model

Our Regression Model employs

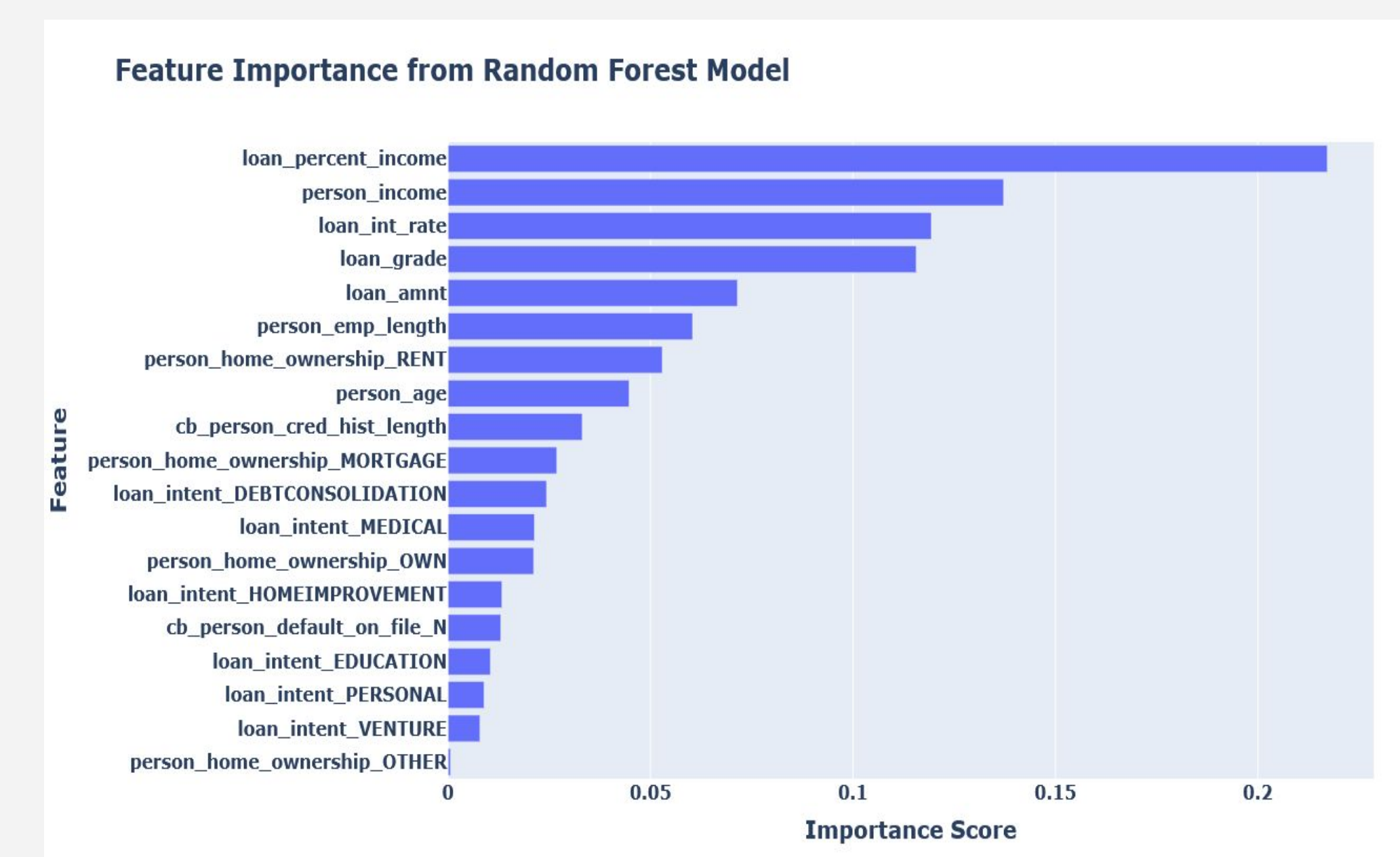
- A 80:20 train-test split
- One Hot Encoding of specific data points like Loan Grade, Loan Intent, and Home Ownership
- Scaling data points to avoid bias
- Accuracy tested against test data to provide user with confidence in the model
 - (Average ~ 85%)



Random Forest Model

Our random forest model employs:

- One-hot encoding for binary decisions
- Columns with similar data were combined or identical data dropped to minimize the number of necessary decisions
- A single forest with 101 trees
 - Multiple forests resulted in overall lower accuracy
- 80:20 train-test split
- Accuracy of ~93% on test data
- Feature importance reinforces correlation data



Results

The Logistic Regression and Random Forest Models are able to correctly predict whether a loan will be approved with an 85% and 93% accuracy, respectively. With the difference in importance of individual variables between the models they provide similar yet still distinct perspectives on the data so that they compliment each other instead of repeating the same information.

Conclusion

The NULL model for the training and testing data set has ~ 78.3% accuracy, which shows a statistical improvement when using our application.